A/B Testing with Exposure-Dependent User Response

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Abstract

Standard A/B-test summaries target a single average effect and therefore obscure how user response changes with repeated exposure to a new product surface. In many applications, it is important to report this dynamic in addition to the average treatment effect: an intervention may initially slow users down and only later improve efficiency, or may exhibit the opposite pattern as novelty wears off. Estimating how response changes over repeated use requires care when users contribute different numbers of exposures within the experiment window. A naive exposure-by-exposure average is confounded by changing user composition.

We study a simple additive model for the conditional mean surface,

$$\mathbb{E}[Y_{ij} \mid T_i = t] = a_0(t) + b_0(j),$$

[Xi: Should I remove the precise model formulation here and replace it with: "We study an additive model in which average outcomes depend on both a user's total number of exposures and the current exposure index."] where Y_{ij} is the outcome at exposure j for user i, T_i is the total number of exposures observed for user i, $a_0(t)$ captures differences across users with different total numbers of exposures, and $b_0(j)$ captures how mean response changes with exposure. We estimate (a_0, b_0) by least squares under the normalization $b_0(1) = 0$. We show that the estimator is conditionally unbiased for b_0, consistent, and asymptotically normal when the exposure horizon is fixed. We further establish first-order validity of the standard user-level bootstrap for inference and show that the naive column-mean estimator generally targets a different limit and can therefore have asymptotically vanishing coverage for the quantity of interest.

CCS Concepts

• **General and reference → Experimentation**.

Keywords

A/B testing, exposure-response curve, semiparametric estimation, clustered bootstrap

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1 Introduction

A product change can affect users differently across repeated exposures. An intervention may attenuate as initial novelty fades, or it may strengthen as users become familiar with a new interface. In such settings, a single average treatment-effect summary can be difficult to interpret and may be inadequate for product decisions. Standard experimentation practice either treats the treatment effect as effectively exposure-invariant or collapses over exposure history into a single summary, and therefore cannot answer rollout questions when user response adapts over repeated exposure.

To motivate the problem, consider an online experiment on a social-networking app. The product team launches a redesigned direct-message reply composer that slightly reduces friction through faster keyboard focus, fewer taps, and quicker attachment access. Heavy and light users may differ substantially in their baseline messaging speed, yet the shape of "getting used to" the new interface may still be similar across user types.

Suppose the team runs an A/B test for two weeks and is interested not only in whether the redesign improves average performance, but also in how user response evolves across repeated uses. Let $T_i \in \{1, \ldots, B\}$ denote the number of direct messages sent by user i during the experiment window, and let $Y_{ij} \in \mathbb{R}$ denote the outcome for the jth exposure, for $j \in [T_i] := \{1, \ldots, T_i\}$. For concreteness, one may take Y_{ij} to be the time spent composing the jth message.

A standard summary is the average within-user outcome,

$$\hat{\theta}_{\text{avg}} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{T_i} \sum_{j=1}^{T_i} Y_{ij},$$

together with a usual z-interval based on the n user-level averages. This is often reasonable for a one-number summary, but it does not isolate how response changes with exposure. If the redesign initially slows users down and only later improves efficiency, then $\hat{\theta}_{\text{avg}}$ conflates these effects.

A natural alternative is the exposure-specific average

$$\left\{ \frac{1}{|\{i : T_i \geq j\}|} \sum_{i:T_i \geq j} Y_{ij} \right\}_{j=1}^{B}.$$

This estimator is generally biased for the exposure profile of interest because the set $\{i : T_i \geq j\}$ changes with j. Later exposures are observed only for users who remain active long enough to reach them, so the estimate increasingly reflects the composition of heavier users.

At the other extreme, one may estimate the full triangular array of cell means,

$$\left\{ \frac{1}{|\{i : T_i = t\}|} \sum_{i:T_i = t} Y_{ij} \right\}_{1 \leq j \leq t \leq B}.$$

This removes the composition problem but introduces $\frac{B(B+1)}{2}$ mean parameters, which can lead to substantial variance.

These observations suggest a bias–variance tradeoff. We would like a model that adjusts for systematic differences across users with different total exposure counts, while avoiding a fully saturated cell-mean surface. Our proposal is a simple additive conditional-mean model on the triangular domain. It yields an exposure-response curve that is interpretable, easy to estimate, and amenable to bootstrap inference.

For clarity, we formulate the method for a single arm. In a two-arm A/B test, one may fit the model separately in each arm and compare the resulting response curves, or apply the same construction to arm-specific contrasts.

2 Related Work

TBD

Our theoretical analysis draws on standard asymptotic arguments for finite-dimensional estimators and on the classical bootstrap. The bootstrap originates with [1]; general large-sample arguments for consistency, asymptotic normality, and the delta method may be found in [3]. Because our data are clustered at the user level, resampling must preserve the within-user dependence structure; the user-level bootstrap is consistent in this setting under standard regularity conditions, and our argument is closely related to the general clustered-data discussion in [2].

3 Setup and Estimand

Fix a maximum exposure horizon $B \in \mathbb{N}$. For each user $i = 1, \dots, n$, we observe a total number of exposures $T_i \in \{1, \dots, B\}$ and outcomes $(Y_{i1}, \dots, Y_{iT_i})$. Define the user record

$$Z_i := (T_i, Y_{i1}, \dots, Y_{iT_i}).$$

We assume throughout that $Z_1, \dots, Z_n$ are i.i.d. across users; within-user dependence in $(Y_{ij})_{j \le T_i}$ is unrestricted.

Let the triangular index set be

$$\mathcal{D} := \{(t, j) : 1 \le j \le t \le B\}.$$

We assume that there exist functions $a_0 : \{1, \dots, B\} \to \mathbb{R}$ and $b_0 : \{1, \dots, B\} \to \mathbb{R}$ such that

$$\mathbb{E}[Y_{ij} \mid T_i = t] = a_0(t) + b_0(j), \qquad (t, j) \in \mathcal{D}, \tag{1}$$

with the normalization

$$b_0(1) = 0. \tag{2}$$

The function $a_0(t)$ captures systematic differences across users who accumulate different total numbers of exposures within the experiment window, while $b_0(j)$ captures the common exposure profile. We refer to $\{b_0(j)\}_{j=1}^{B}$ as the *response curve by exposure*. Under (2), $b_0(j)$ is interpretable as the mean response at exposure j relative to exposure 1, after adjusting for composition through a_0.

Define the regression error

$$\varepsilon_{ij} := Y_{ij} - a_0(T_i) - b_0(j).$$

Then (1) implies

$$\mathbb{E}[\varepsilon_{ij} \mid T_i] = 0 \qquad \text{for all } j \le T_i. \tag{3}$$

[Xi: Explicitely state the estimand.]

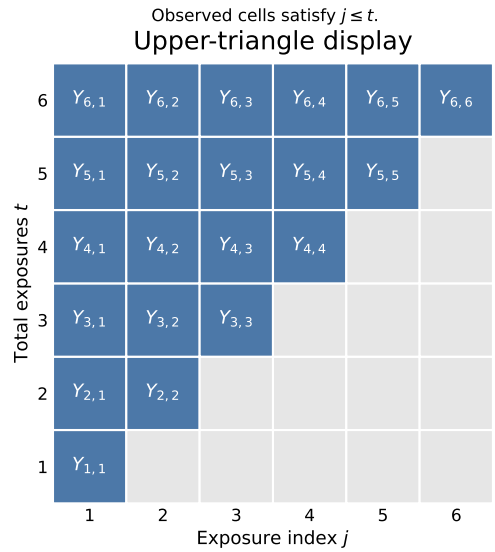

Figure 1: Upper-triangular display of the observed data layout. A cell (t, j) is observed only when $1 \le j \le t \le B$. Under the additive model, the conditional mean at each observed cell satisfies $\mathbb{E}[Y_{ij} \mid T_i = t] = a_0(t) + b_0(j)$.

4 Method

The additive model (1) reduces the $\frac{B(B+1)}{2}$ unrestricted cell means on the triangular domain to $2B - 1$ free parameters. This is the regularization that links the full cell-mean surface to a single exposure curve shifted by a length-specific intercept. It is a strong assumption, but it directly targets the bias–variance tradeoff described in the introduction.

Let

$$\theta := \big(a(1), \dots, a(B),\ b(2), \dots, b(B)\big)^\top \in \mathbb{R}^{2B-1},$$

where $b(1) \equiv 0$ is fixed. For each observed cell (i, j) with $1 \le j \le T_i$, define $x_{ij} \in \mathbb{R}^{2B-1}$ by placing a 1 in the coordinate corresponding to $a(T_i)$, a 1 in the coordinate corresponding to $b(j)$ when $j \ge 2$, and 0 elsewhere. Then

$$x_{ij}^\top \theta = a(T_i) + b(j).$$

We estimate θ by least squares:

$$\hat{\theta} := \arg\min_{\theta \in \mathbb{R}^{2B-1}} \sum_{i=1}^{n} \sum_{j=1}^{T_i} \big(Y_{ij} - x_{ij}^\top \theta\big)^2. \tag{4}$$

Write $\hat{b}(j)$ for the implied estimate of $b_0(j)$, with $\hat{b}(1) = 0$.

For comparison, we also consider the naive column-mean estimator,

$$\hat{b}_{\text{naive}}(j) = \frac{1}{|\{i : T_i \ge j\}|} \sum_{i : T_i \ge j} Y_{ij} - \frac{1}{n} \sum_{i=1}^{n} Y_{i1}, \qquad j = 1, \dots, B,$$

with $\hat{b}_{\text{naive}}(1) = 0$. This estimator ignores the change in user composition across exposure indices.

5 Identification and Finite-Sample Unbiasedness

Lemma 5.1 (Identification under $b_0(1) = 0$). *Assume $\mathbb{P}(T_i = t) > 0$ for all $t \in \{1, \dots, B\}$ and $\mathbb{P}(T_i \geq j) > 0$ for all $j \in \{2, \dots, B\}$. If two pairs (a, b) and (a', b') satisfy*

$$a(t)+b(j) = a'(t)+b'(j) \quad \text{for all } (t, j) \in \mathcal{D}, \qquad b(1) = b'(1) = 0,$$

then $a(t) = a'(t)$ for all t and $b(j) = b'(j)$ for all j.

[Xi: Mention how positivity violation can and should be addressed by coarsening the exposure bins.]

Proof. Fix $t \in \{1, \dots, B\}$. Evaluating the equality at $j = 1$ gives

$$a(t) + b(1) = a'(t) + b'(1),$$

hence $a(t) = a'(t)$ because $b(1) = b'(1) = 0$. Substituting back yields $b(j) = b'(j)$ for all $j \leq t$. Since $\mathbb{P}(T_i \geq j) > 0$ for each j, every exposure index appears in the triangular domain, so $b = b'$ on $\{1, \dots, B\}$. □

Lemma 5.2 (Conditional unbiasedness of $\hat{b}$). *Assume* (1) *and* (3). *Let X be the stacked design matrix whose rows are the vectors $x_{ij}^\top$ over all observed cells (i, j) with $j \leq T_i$. Suppose X has full column rank. Then*

$$\mathbb{E}[\hat{\theta} \mid X] = \theta_0, \qquad \text{and hence} \qquad \mathbb{E}[\hat{b}(j) \mid X] = b_0(j) \quad \text{for all } j \geq 2.$$

Proof. Stack the observed outcomes into a vector Y and write

$$Y = X\theta_0 + \varepsilon,$$

where ε stacks the errors ε_{ij}. The least-squares estimator is

$$\hat{\theta} = (X^\top X)^{-1} X^\top Y = \theta_0 + (X^\top X)^{-1} X^\top \varepsilon.$$

By (3), $\mathbb{E}[\varepsilon \mid X] = 0$, and therefore

$$\mathbb{E}[\hat{\theta} \mid X] = \theta_0.$$

The claim for $\hat{b}(j)$ follows by coordinate projection. □

6 Large-Sample Validity of the Point Estimator

Assumption 6.1 (Regularity conditions for fixed B). *The following conditions hold.*

(1) *$Z_1, \dots, Z_n$ are i.i.d.*
(2) *$T_i \leq B$ almost surely, with fixed B.*
(3) $\mathbb{E}\left[\sum_{j=1}^{T_i} Y_{ij}^2\right] < \infty$.
(4) *The matrix*

$$Q := \mathbb{E}\left[\sum_{j=1}^{T_i} x_{ij} x_{ij}^\top\right]$$

is positive definite.
(5) *The conditional-mean model* (1) *holds.*

Define

$$Q_i := \sum_{j=1}^{T_i} x_{ij} x_{ij}^\top, \qquad q_i := \sum_{j=1}^{T_i} x_{ij} Y_{ij},$$

and the sample averages

$$Q_n := \frac{1}{n} \sum_{i=1}^{n} Q_i, \qquad q_n := \frac{1}{n} \sum_{i=1}^{n} q_i.$$

Whenever Q_n is invertible, the estimator satisfies $\hat{\theta} = Q_n^{-1} q_n$.

[Xi: The abuse of notation in Q_i and Q_n should be avoided. Try something like $\tilde{Q}$ instead.]

Theorem 6.2 (Consistency and asymptotic normality). *Under Assumption 6.1,*

(1) *$\hat{\theta} \xrightarrow{p} \theta_0$, and therefore $\hat{b}(j) \xrightarrow{p} b_0(j)$ for each fixed $j \geq 2$.*
(2) *Let*

$$\psi_i := \sum_{j=1}^{T_i} x_{ij} \varepsilon_{ij}, \qquad \Omega := \mathrm{Var}(\psi_i).$$

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) \Rightarrow N\big(0, Q^{-1} \Omega Q^{-1}\big).$$

Consequently, for each fixed $j \geq 2$,

$$\sqrt{n}\big(\hat{b}(j) - b_0(j)\big) \Rightarrow N(0, v_j),$$

where v_j is the diagonal entry of $Q^{-1} \Omega Q^{-1}$ corresponding to the coordinate $b(j)$.

Proof. By the law of large numbers applied to i.i.d. users, $Q_n \to_p Q$ and $q_n \to_p q := \mathbb{E}[q_i]$. Under (1) and (3),

$$Y_{ij} = x_{ij}^\top \theta_0 + \varepsilon_{ij} \quad \text{with} \quad \mathbb{E}[\varepsilon_{ij} \mid T_i] = 0.$$

Hence

$$q = \mathbb{E}\left[\sum_{j=1}^{T_i} x_{ij}(x_{ij}^\top \theta_0 + \varepsilon_{ij})\right] = Q\theta_0.$$

Because Q is invertible, Q_n is invertible with probability tending to one, and

$$\hat{\theta} = Q_n^{-1} q_n \to_p Q^{-1}(Q\theta_0) = \theta_0.$$

This proves consistency.

For asymptotic normality, observe that

$$Q_n(\hat{\theta} - \theta_0) = q_n - Q_n \theta_0 = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{T_i} x_{ij} \varepsilon_{ij} = \frac{1}{n} \sum_{i=1}^{n} \psi_i.$$

Therefore,

$$\sqrt{n}(\hat{\theta} - \theta_0) = Q_n^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \right).$$

The vectors ψ_i are i.i.d. mean zero with finite covariance Ω, so the multivariate central limit theorem yields

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \Rightarrow N(0, \Omega).$$

Since $Q_n^{-1} \to_p Q^{-1}$, Slutsky's theorem gives the stated limit. The univariate result for $\hat{b}(j)$ follows by projection. See, for example, [3, Chapters 5–6]. □

7 Bootstrap Validity for Our Method

We use the standard nonparametric bootstrap at the user level: draw i.i.d. indices $I_1, \dots, I_n$ uniformly from $\{1, \dots, n\}$, form the bootstrap sample $Z_k^* := Z_{I_k}$, and compute the bootstrap estimator $\hat{\theta}^*$ by solving (4) on the resampled data. Let $\hat{b}^*(j)$ denote the corresponding bootstrap estimate.

Equivalently, if $(N_1, \dots, N_n) \sim \text{Multinomial}(n; 1/n, \dots, 1/n)$ are the bootstrap counts, then $\hat{\theta}^*$ solves

$$\hat{\theta}^* = \arg\min_{\theta \in \mathbb{R}^{2B-1}} \sum_{i=1}^{n} N_i \sum_{j=1}^{T_i} \left(Y_{ij} - x_{ij}^\top \theta\right)^2.$$

Theorem 7.1 (First-order bootstrap validity). *Under Assumption 6.1, for each fixed $j \geq 2$,*

$$\mathcal{L}\left(\sqrt{n}\left(\hat{b}^*(j) - \hat{b}(j)\right) \middle| Z_1, \dots, Z_n\right) \Rightarrow_{\mathbb{P}} \mathcal{L}\left(\sqrt{n}\left(\hat{b}(j) - b_0(j)\right)\right).$$

Consequently, percentile bootstrap confidence intervals for $b_0(j)$ have asymptotically correct first-order coverage.

Proof. The estimator $\hat{\theta}$ is a smooth function of user-level sample averages. Define

$$U_i := \left(\text{vec}(Q_i), q_i\right),$$

so that (Q_n, q_n) is the sample mean of the i.i.d. vectors U_i, and

$$\hat{\theta} = g(Q_n, q_n), \qquad g(Q, q) := Q^{-1} q.$$

On the set where Q is invertible, g is continuously differentiable.

The bootstrap sample (Q_n^*, q_n^*) is the sample mean of i.i.d. draws from the empirical distribution of $\{(Q_i, q_i)\}_{i=1}^n$. Classical bootstrap theory implies that, in fixed dimension and under finite second moments, the conditional law of

$$\sqrt{n}\left((Q_n^*, q_n^*) - (Q_n, q_n)\right)$$

consistently estimates the sampling law of

$$\sqrt{n}\left((Q_n, q_n) - (Q, q)\right);$$

see [1, 3]. Applying the bootstrap delta method to the differentiable map g yields the stated convergence for $\hat{\theta}$, and therefore for the coordinate $\hat{b}(j)$. □

[Xi: Perhaps you want to add some condition so that the probability of Q_n being singular is zero?]

8 The Naive Column-Mean Estimator Targets a Different Quantity

For each j with $\mathbb{P}(T_i \geq j) > 0$, define

$$\bar{Y}_{\cdot j} := \frac{1}{N_j} \sum_{i: T_i \geq j} Y_{ij}, \qquad N_j := \#\{i : T_i \geq j\},$$

and recall

$$\hat{b}_{\text{naive}}(j) = \bar{Y}_{\cdot j} - \bar{Y}_{\cdot 1}, \qquad \hat{b}_{\text{naive}}(1) = 0.$$

Lemma 8.1 (Asymptotic target of the naive estimator). *Assume* (1), (3), *and* $\mathbb{P}(T_i \geq j) > 0$. *Then, for each fixed $j \geq 2$,*

$$\hat{b}_{\text{naive}}(j) \xrightarrow{p} b_0(j) + \Delta(j), \qquad \Delta(j) := \mathbb{E}[a_0(T_i) \mid T_i \geq j] - \mathbb{E}[a_0(T_i)].$$

In particular, if $\Delta(j) \neq 0$, then $\hat{b}_{\text{naive}}(j)$ is inconsistent for $b_0(j)$.

Proof. By the law of large numbers,

$$\bar{Y}_{\cdot j} \to_p \mathbb{E}[Y_{ij} \mid T_i \geq j].$$

Using (1) and (3),

$$\mathbb{E}[Y_{ij} \mid T_i \geq j] = \mathbb{E}[a_0(T_i) + b_0(j) + \varepsilon_{ij} \mid T_i \geq j] = b_0(j) + \mathbb{E}[a_0(T_i) \mid T_i \geq j].$$

Similarly,

$$\bar{Y}_{\cdot 1} \to_p \mathbb{E}[Y_{i1}] = b_0(1) + \mathbb{E}[a_0(T_i)] = \mathbb{E}[a_0(T_i)].$$

Subtracting the two limits yields the result. □

Theorem 8.2 (Failure of naive *i.i.d.* intervals for $b_0(j)$). *Fix $j \geq 2$ and suppose $\Delta(j) \neq 0$. Consider any confidence interval of the form*

$$\text{CI}_{\text{naive}}(j) = \left[\hat{b}_{\text{naive}}(j) - c_{n,j}, \hat{b}_{\text{naive}}(j) + c_{n,j}\right], \qquad c_{n,j} = O_p(n^{-1/2}),$$

which includes the usual normal or t intervals formed under an i.i.d. assumption. Then

$$\mathbb{P}\left(b_0(j) \in \text{CI}_{\text{naive}}(j)\right) \to 0.$$

Proof. By Lemma 8.1,

$$\hat{b}_{\text{naive}}(j) \to_p b_0(j) + \Delta(j), \qquad \Delta(j) \neq 0.$$

Therefore,

$$\hat{b}_{\text{naive}}(j) - b_0(j) \to_p \Delta(j),$$

while $c_{n,j} \to_p 0$. It follows that

$$\mathbb{P}\left(\left|\hat{b}_{\text{naive}}(j) - b_0(j)\right| \leq c_{n,j}\right) \to 0,$$

which is equivalent to

$$\mathbb{P}\left(b_0(j) \in \text{CI}_{\text{naive}}(j)\right) \to 0.$$

□

9 Experiments

We study a finite-sample design that matches the model in (1). For each replication, we first draw T_i from a truncated geometric distribution on $\{1, \dots, B\}$, with mean approximately $(B+1)/2$ before truncation. Conditional on T_i, outcomes follow

$$Y_{ij} = a_0(T_i) + b_0(j) + u_i + \varepsilon_{ij}, \qquad j \leq T_i,$$

where

$$a_0(t) = 0.25 \log(1+t), \qquad b_0(j) = -0.6\left(1 - e^{-(j-1)/6}\right), \qquad b_0(1) = 0.$$

The user effect satisfies $u_i \sim N(0, \tau_u^2)$, and the within-user noise is stationary AR(1):

$$\varepsilon_{ij} = \rho\, \varepsilon_{i,j-1} + \eta_{ij}, \qquad \eta_{ij} \sim N(0, \sigma_\eta^2).$$

We calibrate the noise level through

$$\text{SNR} := \frac{\max_j b_0(j) - \min_j b_0(j)}{\text{sd}(u_i + \varepsilon_{ij})}, \qquad \text{ICC} := \frac{\text{Var}(u_i)}{\text{Var}(u_i + \varepsilon_{ij})}.$$

Throughout, we fix ICC $= 0.2$ and consider $B \in \{5, 30\}$, $n = 1000$, SNR $\in \{0.1, 0.2, 0.3\}$, and $\rho \in \{0.3, 0.6, 0.8\}$. This yields 18 scenarios. For each scenario we run 200 Monte Carlo replications. In each replication, we compute our least-squares estimator together with a percentile user-level bootstrap interval based on 200 bootstrap resamples, and we compare it with the naive column-mean estimator paired with the usual normal interval that treats the exposure-specific means as if they were i.i.d.

We report pointwise bias, standard deviation, mean squared error, and 95% coverage for $b_0(j)$, together with scenario-level summaries obtained by averaging over $j \geq 2$. Table 1 reports the main aggregate comparison. Across both horizons, the proposed estimator is nearly unbiased, whereas the naive estimator exhibits substantial positive bias induced by composition changes across

Table 1: Scenario-averaged performance over $j \geq 2$. Each row averages the scenario-level summaries over the nine combinations of SNR and ρ for a fixed horizon B.

B	Method	Avg. sd	Avg. |bias|	Avg. MSE	Avg. cov.
5	Ours	0.090	0.0049	0.0108	0.942
5	Naive	0.096	0.1136	0.0259	0.771
30	Ours	0.211	0.0120	0.0579	0.942
30	Naive	0.209	0.1965	0.0999	0.819

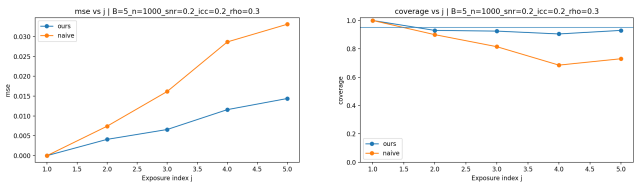

Figure 2: Representative finite-sample behavior for $B = 5$, $n = 1000$, SNR $= 0.2$, ICC $= 0.2$, and $\rho = 0.3$. Left: mean squared error by exposure index. Right: pointwise 95% coverage, with the horizontal line marking nominal coverage.

exposure indices. The resulting MSE gains are sizeable: the average MSE is reduced from 0.0259 to 0.0108 when $B = 5$ and from 0.0999 to 0.0579 when $B = 30$. The bootstrap interval is also much better calibrated, with average coverage close to nominal in both horizons.

Figure 2 shows a representative scenario with $B = 5$, SNR $=$ 0.2, and $\rho = 0.3$. The MSE gap widens with the exposure index, even though the two estimators have comparable dispersion. This reflects the main mechanism in the simulations: our estimator gains primarily by removing the composition bias that accumulates at later exposures. The same pattern appears in the coverage curves. The bootstrap interval remains close to the 95% target, while the naive interval undercovers markedly once j becomes moderate. Across the full grid, the average coverage of our interval ranges from 0.919 to 0.960, whereas the corresponding range for the naive interval is 0.556 to 0.951; at individual exposure indices, the naive coverage can fall as low as 0.40.

Two features are worth emphasizing. First, the naive estimator can look competitive in standard deviation alone, and in some scenarios it is slightly less variable. Nevertheless, its persistent composition bias dominates the risk comparison, so its MSE remains uniformly larger in our experiments. Second, the difference between the methods is most pronounced when the signal is stronger relative to the noise and when the exposure index is large. This is exactly the regime in which the theoretical distinction between $b_0(j)$ and the naive target is most consequential.

10 Conclusion

We study estimation of an exposure-response curve in triangular panel data arising in online experimentation. The proposed additive conditional-mean model yields a simple estimator that adjusts for composition across users with different total exposure counts while retaining a parsimonious representation of the mean surface. Under the model, the estimator is unbiased and asymptotically normal, and the standard user-level bootstrap is valid for inference. In contrast, the naive exposure-wise average generally targets a different limit and can have asymptotically vanishing coverage for the curve of interest.

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